

K-FIELD

RECURSIVE LUCAS SPECTRUM

Exact scale recursion, morphological eigenchannels, harmonic and subharmonic band formation, reciprocal-mode nesting, and arbitrary-volume closure laws

Recursive Lucas spectrum: geometry, scaling, and exact subharmonic folding

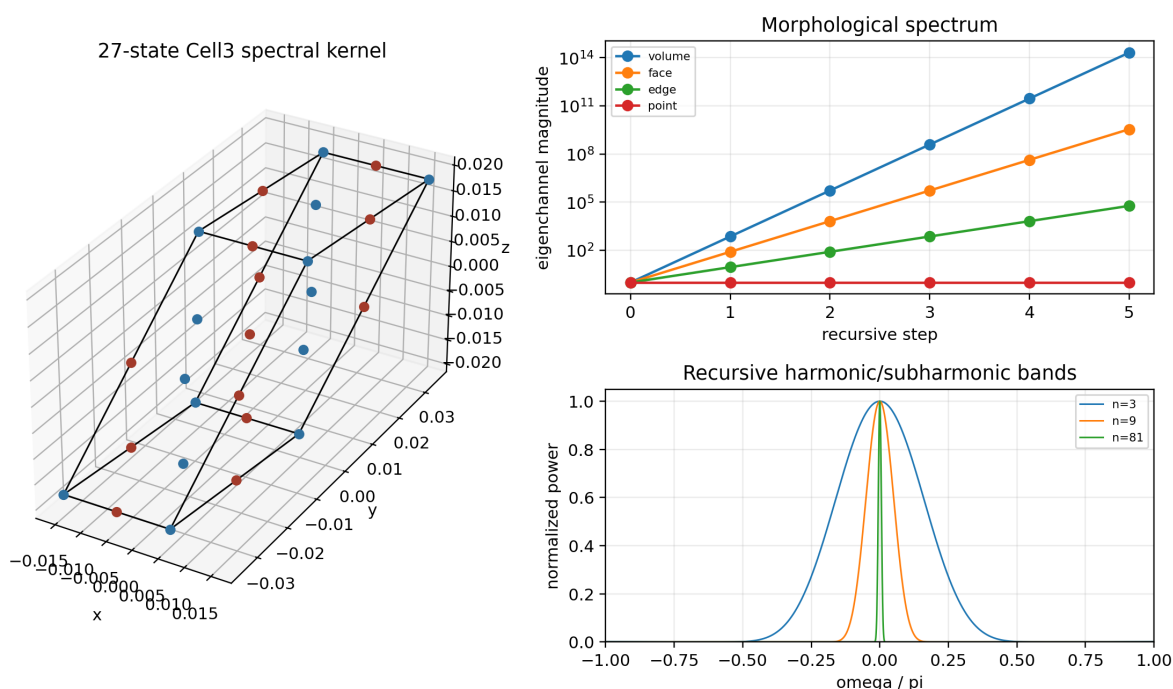


Figure 1. Title composite generated entirely from the authoritative Cell3 basis, exact recursive scaling channels, and numerically evaluated Dirichlet spectra.

Principal Investigator

P. Dwayne Esterline

BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008) | Krytronx

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Archival computational and educational edition - Unix epoch 1785243336 - 2026-07-28T12:55:36+00:00

Document Control

Field	Value
Title	K-Field Recursive Lucas Spectrum
Release type	Foundational educational and archival computational reference
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx
Epoch identifier	1785243336
UTC generation time	2026-07-28T12:55:36+00:00
Classification	PROPRIETARY CONFIDENTIAL INFORMATION AND TRADE SECRET OF KRYTRONX, LLC. All Rights Reserved. Do Not Distribute.
Coordinate convention	Canonical right-handed Cartesian reconstruction: a on +x, b in the x-y plane, positive-z branch for c
Native units	Native geometric length, area, volume, and reciprocal-length units unless explicitly normalized
Machine-readable companion	K-Field_recursive_Lucas_Spectrum1785243336.json
Primary internal reference format	K-Field_Lucas_Supergroup_Complete_Nuclear_Closure_Domain_1785235808.pdf

Central result. The Lucas Supergroup is the exact 9B oblique cubical complex containing 729 Cell3 volumes and 27 Lucas Groups. A 9 x 9 x 9 array of Lucas Supergroups is an exact second-order Lucas closure. It reproduces the complete Lucas-Supergroup census at the Lucas-Supergroup scale while containing 729 Lucas Supergroups, 19,683 Lucas Groups, and 531,441 Cell3 volumes.

Contents

Document Control	2
Abstract	5
1. Reader Orientation and Scale Conventions	6
2. Authoritative Cell3 Geometry	6
2.1 Direct and Reciprocal Metrics	7
2.2 Primitive Planes, Face Areas, and Volume	8
3. The Lucas Supergroup as an Exact 9B Complex	8
3.1 Exact Second-Order Lucas Closure	9
4. Exhaustive Aligned Volume-Filling Law	9
5. Recursive Lucas Closures of Arbitrary Order	10
6. Exact Morphological Scaling Spectrum	11
7. Complete Invariant Spectrum	12
8. The Seven-Sheet Cell3 Kernel	13
9. Fourier Harmonics and Subharmonics	14
10. Exact 729-Band Folding	15
11. Reciprocal-Space Mode Nesting	17
12. Unique Recursive Factorization of Arbitrary Volumes	19
13. Recursive Energy-Band Structure	20
14. Universal Four-Channel Recurrence	21
15. Cycle Space and Path-Entropy Spectrum	22
16. New Recursive Spectral Properties	22
16.1 Exact discrete scale invariance	22
16.2 Two-stage ternary spectral birth	23
16.3 Exact mode inheritance and spectral-gap contraction	23
16.4 Fixed band-birth census	23

16.5 Noncentral polarity cancellation	23
16.6 Scale-invariant anisotropic splitting	23
16.7 Boundary-renormalization law	23
16.8 Dimensional crossover in anisotropic arrays	23
16.9 Finite-difference and recurrence spectroscopy	23
16.10 Closure selection rules	23
17. Computational and Experimental Examination Protocol	23
17.1 Geometry and census test	24
17.2 Laminar polynomial test	24
17.3 Fourier band test	24
17.4 Reciprocal metric test	24
17.5 Residue-sector test	24
17.6 Energy and route-space test	24
18. Integrated Interpretation	24
Appendix A. Formula and Theorem Ledger	25
Appendix B. Complete 27-Node Cell3 Ledger	26
Appendix C. Recursive Closure Ledger	27
C.1 Cellular census by recursive order	27
C.2 Refined-state, boundary, and interface census	27
C.3 Recursive hierarchy and laminar-cycle census	27
Appendix D. Lowest Reciprocal Modes	28
Appendix E. Machine-Readable Companion	30
References and Source Record	30

Abstract

The Lucas Supergroup is an exact finite oblique cubical complex whose primitive census, refined state lattice, interface structure, laminar decomposition, and independent cycle space are all fixed by a nine-cell linear closure. Its complete Cell3-resolution census is $(f_0, f_1, f_2, f_3) = (1000, 2700, 2430, 729)$. The same census reappears at the Lucas-Supergroup scale when $9 \times 9 \times 9$ Lucas Supergroups are assembled. This equality is an exact change of scale: vertices at the coarse scale correspond to complete lower-level objects, edges correspond to face-sharing chains of those objects, faces correspond to two-dimensional arrays, and volumes correspond to three-dimensional arrays.

The recursive hierarchy has a discrete scale factor of 9 in length. Every local cubical observable therefore separates into four exact scaling channels: a volume channel multiplied by 729, a face channel multiplied by 81, an edge channel multiplied by 9, and a point channel multiplied by 1. These channels form the morphological spectrum $\{729, 81, 9, 1\}$. Relative to the bulk term, surface, edge, and point corrections contract as $1/9$, $1/81$, and $1/729$ per closure.

The same recursion generates spatial harmonics and subharmonics. The seven-sheet Cell3 polynomial $\Lambda_3(z) = (1+z+z^2)^3$ is the irreducible laminar kernel. Because $9=3^2$, every Lucas closure introduces two consecutive ternary spectral generations. At each scale a parent reciprocal mode folds into 729 residue sectors: one inherited sector, 26 Lucas-intermediate sectors, and 702 closure-native sectors. Noncentral sectors are exactly polarity balanced. The oblique reciprocal metric then splits the recursively generated mode families into direction-dependent eigenvalues while preserving exact scale inheritance.

This paper develops the geometry, census laws, recursive generating functions, Fourier spectra, residue-class folding, reciprocal-mode nesting, energy-band recurrence, cycle-space scaling, arbitrary-volume factorization, and the resulting new spectral properties as one internally complete reference.

Keywords: K-Field, Cell3, Lucas Group, Lucas Supergroup, exact recursion, discrete scale invariance, Dirichlet kernel, spectral folding, oblique lattice, reciprocal metric, harmonic bands, subharmonics, cycle rank.

1. Reader Orientation and Scale Conventions

A cubical complex is a collection of points, line segments, faces, and three-dimensional cells joined without gaps or overlaps. The word cubical describes the incidence pattern, not an orthogonal metric. In the K-Field hierarchy the primitive Cell3 is oblique: its three basis vectors have unequal lengths and non-right mutual angles. The topology is cubical while the metric is anisotropic.

Two resolutions must be kept distinct. At absolute Cell3 resolution, one Lucas Supergroup contains 9 cells along each basis direction. At Lucas-Supergroup resolution, the same physical object is one coarse three-cell. A $9 \times 9 \times 9$ array of these coarse cells has the same incidence census as one Lucas Supergroup had at Cell3 resolution. Exact recursion means that the complete algebraic census is preserved after this relabeling of scale.

$$\text{Absolute closure order } L_r: n_r = 9^r \text{ Cell3 per axis, } r \geq 1. \\ N_{\text{Cell3}}(r) = 729^r; N_{\text{SG}}(r) = 729^{(r-1)}; N_{\text{LG}}(r) = 27 * 729^{(r-1)}.$$

The letter n denotes a number of primitive cells per axis at the resolution under discussion. The letters a, b, and c denote whole-number counts of aligned Lucas Supergroups along the three authoritative basis directions. Bold or matrix B denotes the ordered Cell3 basis, $B = [a \ b \ c]$; the meaning is always clear from context.

The recursion studied here is exact for aligned arrays. Arbitrary aligned rectangular arrays are allowed, but full three-dimensional closure occurs only when each axis contains the required radix-9 factor. Different radix-9 depths along the three axes produce anisotropic recursive envelopes rather than complete isotropic closure.

2. Authoritative Cell3 Geometry

The primitive metric is fixed by three edge lengths and three included angles. A dimensionally faithful Cartesian reconstruction places vector a on the positive x axis, vector b in the x-y plane, and vector c on the positive-z branch. This construction changes only the viewing frame; it preserves every length, angle, area, volume, direct metric entry, and reciprocal metric entry.

$$a = (|a|, 0, 0) \\ b = (|b| \cos(\theta_{ab}), |b| \sin(\theta_{ab}), 0) \\ c_x = |c| \cos(\theta_{ac}) \\ c_y = (|b| |c| \cos(\theta_{bc}) - b_x c_x) / b_y \\ c_z = \sqrt{(|c|^2 - c_x^2 - c_y^2)}.$$

Quantity	Value	Units / meaning
a	0.0207970587	native length
b	0.0364942052	native length
c	0.0493085660	native length
theta_ab	83.933483	degrees
theta_ac	80.448130	degrees
theta_bc	50.363232	degrees
a	[0.0207970587, 0.0, 0.0]	canonical Cartesian vector
b	[0.00385681612, 0.036289833047, 0.0]	canonical Cartesian vector
c	[0.008182284384, 0.03076238238, 0.037657147177]	canonical Cartesian vector
det(B)	2.842066945211215e-05	native volume, signed
det(B)	2.842066945211215e-05	Cell3 volume

Authoritative Cell3 geometry and complete signed-address atlas

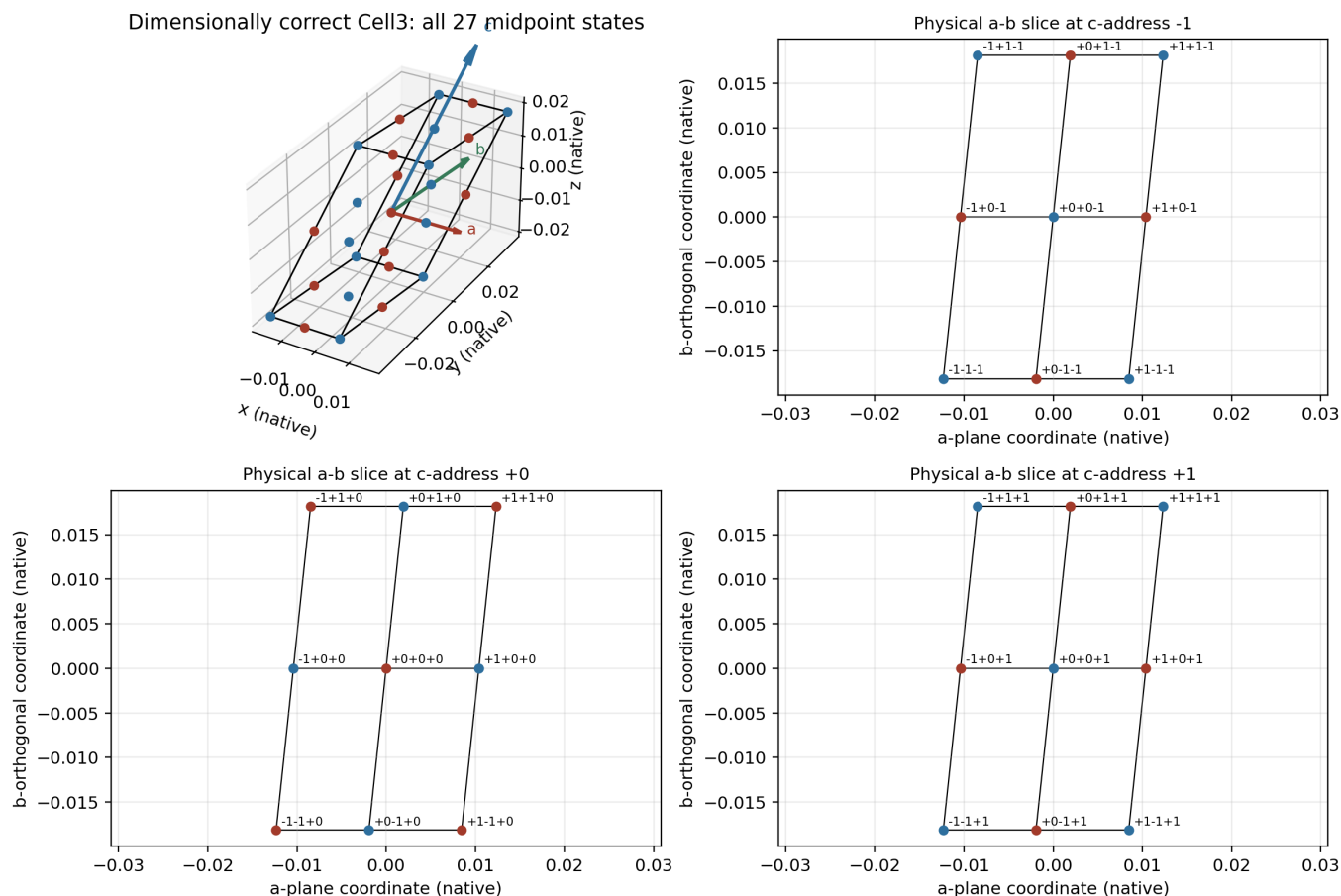


Figure 2. Dimensionally correct Cell3 rendering. Every one of the 27 midpoint-address states is plotted. The three physical a-b plane slices label all signed addresses without projection degeneracy; the perspective panel shows the 12 enclosing edges and ordered basis vectors. Red and blue encode the two checkerboard polarities $\chi=(-1)^\lambda$.

2.1 Direct and Reciprocal Metrics

The direct metric $M=B^T B$ contains the squared lengths on its diagonal and the pairwise dot products off the diagonal. The reciprocal metric $R=M^{-1}$ converts integer mode addresses into inverse-square spectral values. Direct and reciprocal geometry are dual descriptions of the same Cell3.

$$M=B^T B, \quad R=M^{-1}, \quad q(n)=n^T R n.$$

M row	a	b	c
a	4.325176505712456e-04	8.021043124313843e-05	1.701674486329727e-04
b	8.021043124313843e-05	1.331827013179707e-03	1.147919287031111e-03
c	1.701674486329727e-04	1.147919287031111e-03	2.431334680976355e-03
R row	a*	b*	c*
a*	2377.512203001113	0.396285007181	-166.587550471955
b*	0.396285007181	1266.057428561365	-597.778326001406
c*	-166.587550471955	-597.778326001406	705.188414264506

The off-diagonal reciprocal entries retain the exchange and interference structure generated by the oblique direct basis. Scaling the cell by a factor L multiplies M by L^2 and divides R by L^2 . This simple metric duality is the source of the exact $1/81$ reciprocal contraction under a length factor of 9.

2.2 Primitive Planes, Face Areas, and Volume

Each pair of basis vectors spans a primitive parallelogram. The face area is the norm of the corresponding cross product. These three unequal areas make the Cell3 directionally structured even though its incidence pattern is cubical.

$$A_{ab}=|a \times b|, A_{ac}=|a \times c|, A_{bc}=|b \times c|, V=|a \cdot (b \times c)|.$$

Plane	Spanning vectors	Included angle	Computed area	Registered area	Absolute difference
ab	a and b	83.933483 deg	7.547217880965119e-04	7.547217880965119e-04	0.000e+00
ac	a and c	80.448130 deg	1.011255755679060e-03	1.011255755679060e-03	0.000e+00
bc	b and c	50.363232 deg	1.385784440909538e-03	1.385784440909538e-03	2.168e-19

Three primitive Cell3 planes at one common dimensional scale

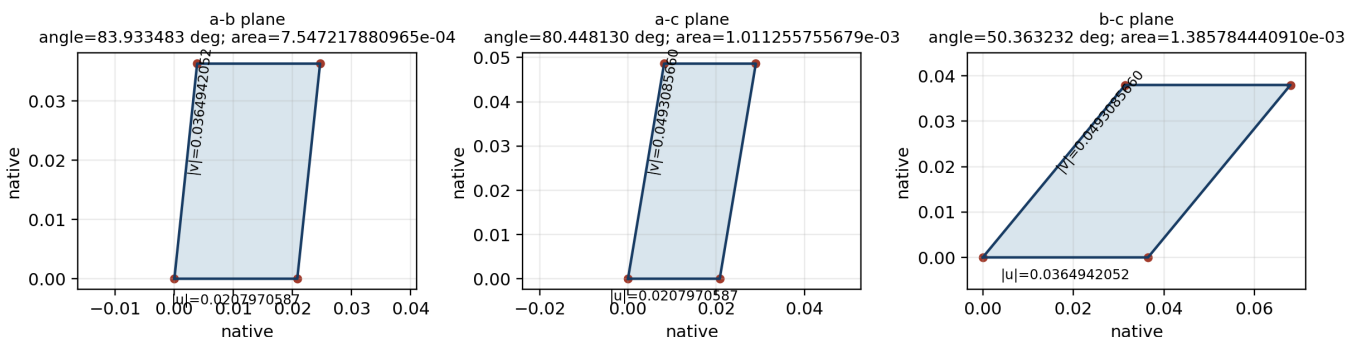


Figure 3. The three primitive Cell3 planes drawn at one common dimensional scale. The unequal edge lengths and included angles are preserved; each listed area is computed from the reconstructed basis vectors.

The 27 local midpoint states are located at $p(s)=(1/2)B s$ with s in $\{-1,0,1\}^3$. Their laminar grade is $\lambda=s_1+s_2+s_3$, and their checkerboard polarity is $\chi=(-1)^\lambda$. The eight states with no zero address components are the Cell3 vertices, the twelve with one zero component are edge midpoints, the six with two zero components are face-center states, and the all-zero address is the center.

3. The Lucas Supergroup as an Exact 9B Complex

The Lucas Supergroup occupies nine Cell3 intervals along each ordered basis direction. Its outer basis is 9B. Because an $n \times n \times n$ cubical block contains $(n+1)^3$ corner nodes, $3n(n+1)^2$ edges, $3n^2(n+1)$ faces, and n^3 volumes, substitution of $n=9$ gives the defining census.

$$(f_0, f_1, f_2, f_3) = ((n+1)^3, 3n(n+1)^2, 3n^2(n+1), n^3).$$

At $n=9$: $(f_0, f_1, f_2, f_3) = (1000, 2700, 2430, 729).$

Object or invariant	Exact value	Construction
Cell3 volumes	729	9^3
Lucas Groups	27	3^3
Unrefined corner nodes	1000	10^3
Unrefined edges	2700	$3 \cdot 9 \cdot 10^2$
Unrefined faces	2430	$3 \cdot 9^2 \cdot 10$
Refined barycentric states	6859	19^3
Laminar sheets	55	$3 \cdot 19 - 2 = 6 \cdot 9 + 1$
Internal Cell3 interfaces	1944	$3 \cdot 9^2 \cdot (9-1)$
Boundary states	1946	$24 \cdot 9^2 + 2$
Exposed primitive faces	486	$6 \cdot 9^2$
Center-graph cycle rank	1216	$2 \cdot 9^3 - 3 \cdot 9^2 + 1$

Object or invariant	Exact value	Construction
Central laminar sheet states	271	$3^9 \cdot 2 + 3^9 + 1$

Interface-boundary identity. At $n=9$ only, the refined boundary-state count equals the internal interface count plus the Euler characteristic of a sphere: $1946=1944+2$. This is an independent exact selection of the nine-cell closure scale.

Exact Lucas Supergroup populations in the locked oblique basis

1000 unrefined corner nodes ($10 \times 10 \times 10$)

6859 refined barycentric states ($19 \times 19 \times 19$)

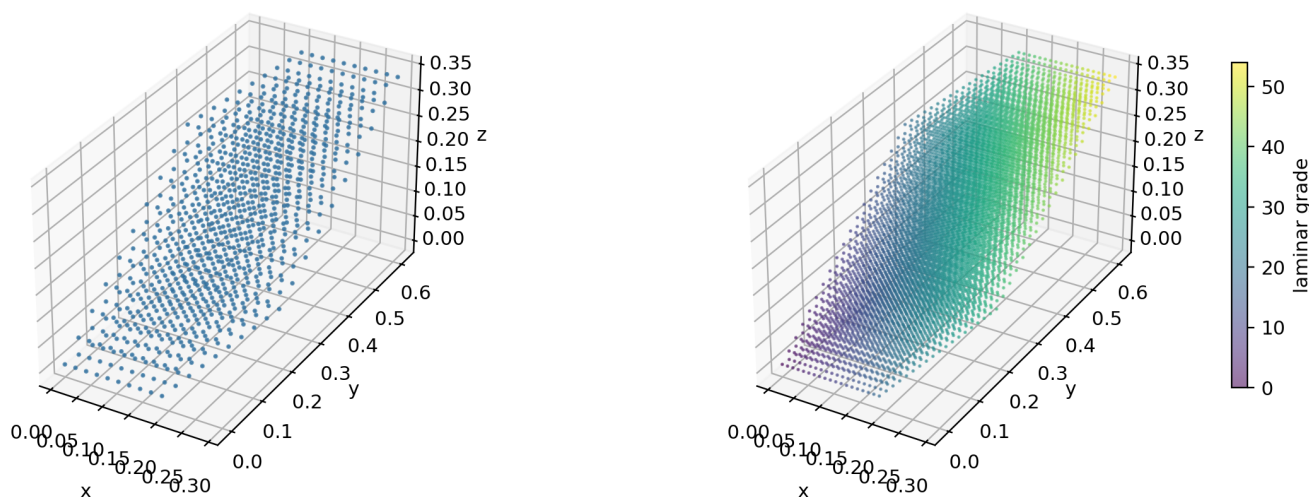


Figure 4. Complete Lucas-Supergroup populations in the dimensionally correct oblique basis. Left: all 1000 unrefined corner nodes. Right: all 6859 refined barycentric states, colored by laminar grade.

3.1 Exact Second-Order Lucas Closure

A $9 \times 9 \times 9$ array of Lucas Supergroups contains 729 lower-level Supergroups. At the coarse Supergroup scale it has ten coarse vertices per axis, nine coarse edges per axis direction, and the complete census (1000,2700,2430,729). This is exactly the same census that one Lucas Supergroup has at Cell3 resolution.

$$729 \text{ Lucas Supergroups} = 729 \cdot 27 \text{ Lucas Groups} = 729 \cdot 729 \text{ Cell3 volumes} \\ = 19,683 \text{ Lucas Groups} = 531,441 \text{ Cell3 volumes.}$$

Resolution	Cell count per axis	f0	f1	f2	f3
One Lucas Supergroup at Cell3 scale	9	1000	2700	2430	729
$9 \times 9 \times 9$ Supergroups at SG scale	9	1000	2700	2430	729
Same second-order object at Cell3 scale	81	551368	1633932	1614006	531441

The second row and third row are identical because the recursive map replaces each coarse cell by a complete lower-level Lucas Supergroup without changing the incidence rule. The fourth row is the absolute fine-resolution census of the same physical aggregate. Exact recursion therefore means equality after scale renormalization, not equality between fine and coarse raw counts.

4. Exhaustive Aligned Volume-Filling Law

Let an aligned rectangular aggregate contain a $x \times b \times c$ Lucas Supergroups. Alignment means that every component uses the same ordered basis B and that neighboring Supergroups share complete faces. The outer basis is

$$B_{\text{out}} = 9 \ B \ \text{diag}(a, b, c).$$

The populations of Lucas Supergroups, Lucas Groups, and Cell3 volumes are

$$N_SG=abc, \quad N_LG=27abc, \quad N_Cell3=729abc.$$

At the Lucas-Supergroup scale, one interval along an axis contributes one coarse edge and two coarse vertices. The complete generating polynomial is

$$F_{\{a,b,c\}}(t)=[(a+1)+at][(b+1)+bt][(c+1)+ct].$$

The coefficient of t^d is the number of d -dimensional cells at that resolution. Expansion gives

$$\begin{aligned} f_0 &= (a+1)(b+1)(c+1) \\ f_1 &= a(b+1)(c+1) + (a+1)b(c+1) + (a+1)(b+1)c \\ f_2 &= ab(c+1) + a(b+1)c + (a+1)bc \\ f_3 &= abc. \end{aligned}$$

Refining every coarse interval by its midpoint produces $(2a+1)(2b+1)(2c+1)$ coarse-scale barycentric states. The remaining exact invariants are

$$\begin{aligned} Q_SG &= (2a+1)(2b+1)(2c+1) \\ B_SG &= 8(ab+ac+bc)+2 \\ I_SG &= 3abc - (ab+ac+bc) \\ E_ext &= 2(ab+ac+bc) \\ \beta_1 &= 2abc - (ab+ac+bc)+1. \end{aligned}$$

At Cell3 resolution substitute $p=9a$, $q=9b$, and $r=9c$. This yields

$$\begin{aligned} Q_Cell3 &= (18a+1)(18b+1)(18c+1) \\ B_Cell3 &= 648(ab+ac+bc)+2 \\ I_Cell3 &= 2187abc - 81(ab+ac+bc) \\ E_Cell3, ext &= 162(ab+ac+bc). \end{aligned}$$

These equations exhaust every aligned rectangular volume-filling set because every interior interface is shared by exactly two neighboring cells and every boundary face belongs to exactly one cell. No approximation enters the census.

$a \times b \times c$	N_SG	N_LG	N_Cell3	Q_SG	B_SG	I_SG	β_1
$1 \times 1 \times 1$	1	27	729	27	26	0	0
$2 \times 3 \times 5$	30	810	21870	385	250	59	30
$9 \times 9 \times 9$	729	19683	531441	6859	1946	1944	1216
$18 \times 45 \times 72$	58320	1574640	42515280	488215	42770	169614	111295
$81 \times 27 \times 9$	19683	531441	14348907	170335	25274	55890	36208

5. Recursive Lucas Closures of Arbitrary Order

Let L_r denote the absolute closure of order r , with L_1 equal to one Lucas Supergroup. The number of Cell3 intervals per axis is $n_r=9^r$. Length, area, and volume therefore scale by 9, 81, and 729 at every transition L_r to L_{r+1} .

$$\begin{aligned} n_r &= 9^r \\ N_Cell3(r) &= 9^r(3r) = 729^r \\ N_SG(r) &= 729^{(r-1)} \\ N_LG(r) &= 27 \cdot 729^{(r-1)}. \end{aligned}$$

r	Cell3/axis	Lucas SG	Lucas Groups	Cell3 volumes	f_0	Q	boundary	interfaces	β_1
1	9	1	27	729	1000	6859	1946	1944	1216
2	81	729	19683	531441	551368	4330747	157466	1574640	1043200
3	729	531441	14348907	387420489	389017000	3105745579	12754586	1160667144	773246656
4	6561	387420489	10460353203	282429536481	282558696328	2259952891867	1033121306	847159469280	564729932800
5	59049	282429536481	762559748498 7	205891132094649	205901592625 000	164717089852429 9	83682825626	6176629359307 44	411771803836096

The integers grow rapidly, but their algebraic structure remains low dimensional. Every count is a polynomial in n_r , and n_r itself is a single exponential 9^r . The recursion can therefore be diagonalized into a small number of exact scaling channels.

6. Exact Morphological Scaling Spectrum

Consider an arbitrary aligned base shape $a \times b \times c$ and multiply each linear dimension by 9^r at the chosen resolution. Its generating polynomial is

$$F^r(a,b,c)(t) = \text{product over } \mu \text{ in } \{a,b,c\} \text{ of } [1 + 9^\mu t^\mu].$$

Expanding by geometric dimension gives

$$F^r(a,b,c)(t) = 729^r abc(1+t)^3 + 81^r (ab+ac+bc)(1+t)^2 + 9^r (a+b+c)(1+t) + 1.$$

The four multipliers are the eigenvalues of the recursive morphology operator:

Morphological scaling spectrum: {729, 81, 9, 1}. The associated eigenchannels are volume, face/interface, edge/line, and point/Euler content.

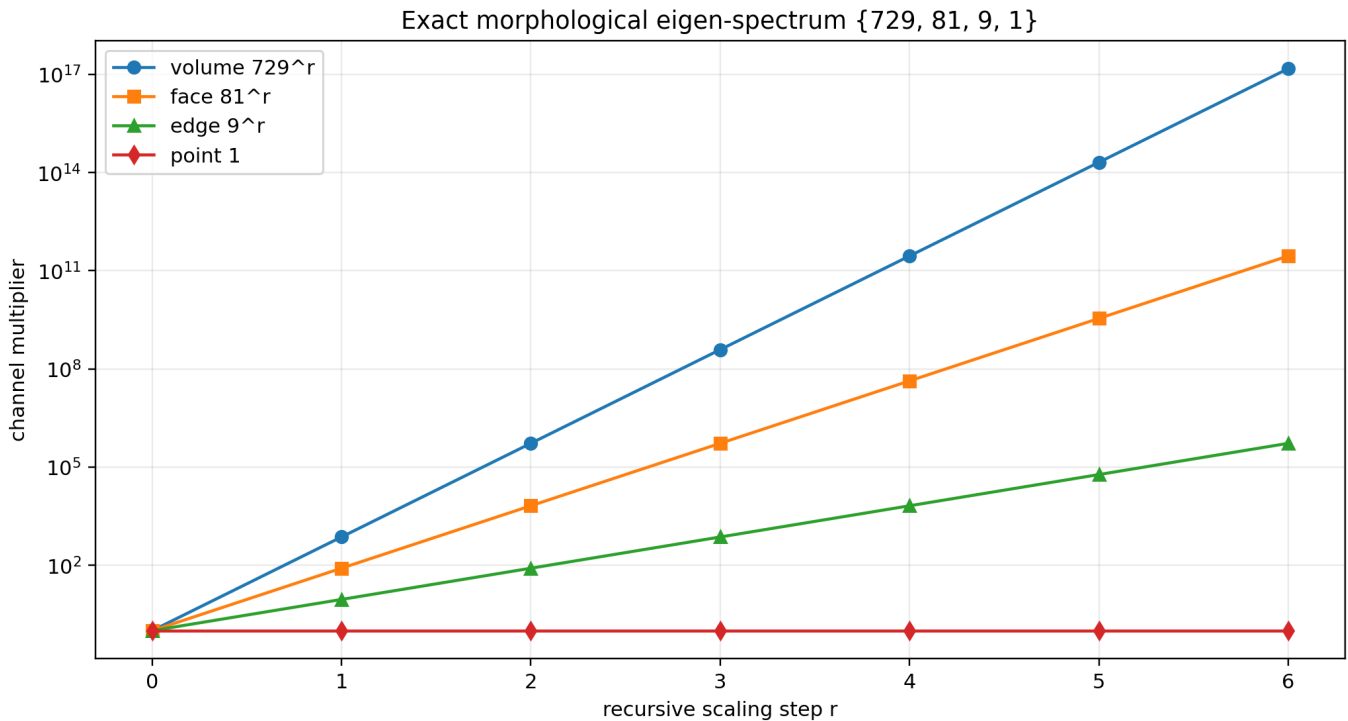


Figure 5. Exact recursive growth of the four morphological eigenchannels. The vertical axis is logarithmic; adjacent channels differ by one factor of 9 in their effective dimension.

For the canonical closure $a=b=c=1$, the polynomial becomes

$$F_r(t) = 729^r (1+t)^3 + 3 \cdot 81^r (1+t)^2 + 3 \cdot 9^r (1+t) + 1.$$

At $r=1$ this is $(10+9t)^3$, which expands to the Lucas-Supergroup census. Evaluation at $t=-1$ eliminates every positive-dimensional term:

$$F_r(-1) = 1.$$

The fixed value 1 is the Euler characteristic of a contractible three-dimensional block. It is the point-channel eigenmode of the recursive operator. Relative to the volume channel, the lower-dimensional terms scale as

volume: 1, face: $1/9^r$, edge: $1/81^r$, point: $1/729^r$.

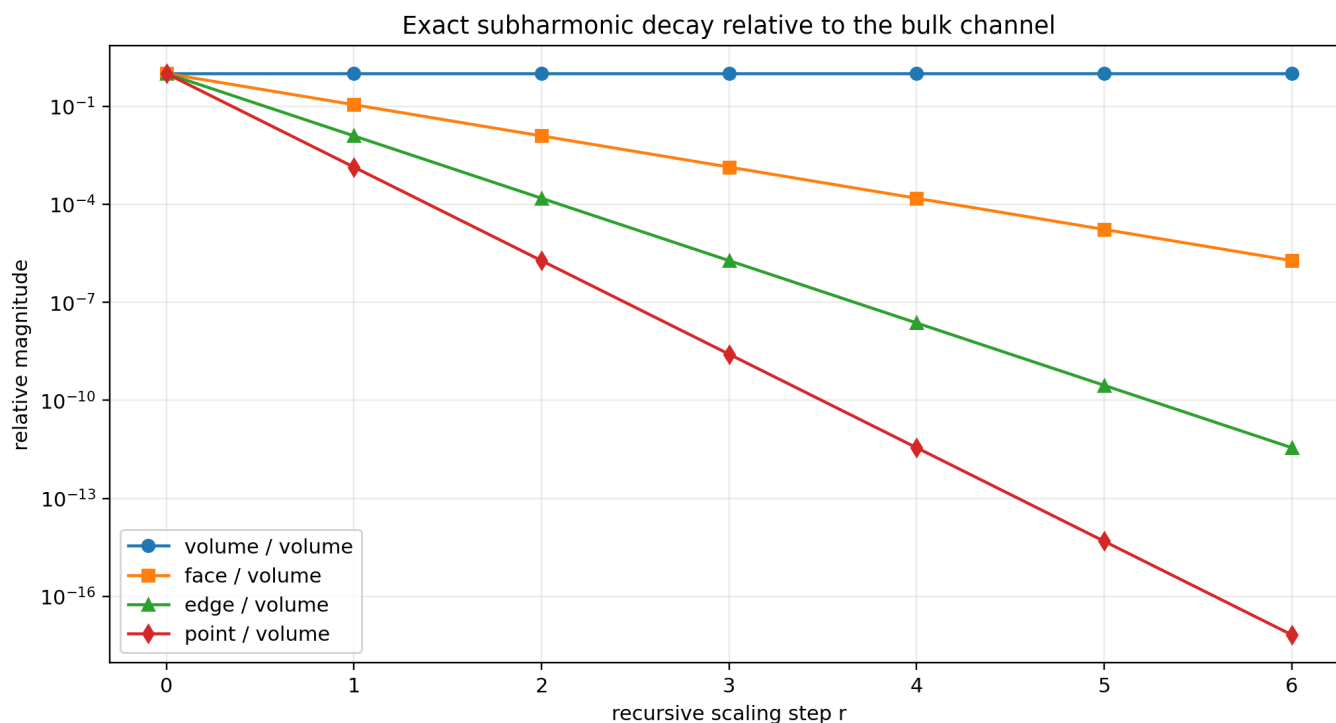


Figure 6. Exact lower-dimensional subharmonics relative to the bulk channel. Every recursive closure suppresses the face correction by $1/9$, the edge correction by $1/81$, and the point correction by $1/729$.

These are morphological subharmonics because they are lower powers of the same fundamental linear scale. They are not fitting residuals. A finite aggregate always contains all four channels; large aggregates become bulk dominated because the lower-dimensional channels contract relative to volume.

7. Complete Invariant Spectrum

The principal Lucas invariants are polynomials in n_r and therefore decompose into the same eigenchannels. Substituting $n_r=9^r$ gives

$$\begin{aligned}
 Q_r &= (2n_r+1)^3 = 8 \cdot 729^r + 12 \cdot 81^r + 6 \cdot 9^r + 1 \\
 B_r &= 24n_r^2 + 2 = 24 \cdot 81^r + 2 \\
 I_r &= 3n_r^2(n_r-1) = 3 \cdot 729^r - 3 \cdot 81^r \\
 E_{\text{ext},r} &= 6n_r^2 = 6 \cdot 81^r \\
 \text{beta}_{1,r} &= 2n_r^3 - 3n_r^2 + 1 = 2 \cdot 729^r - 3 \cdot 81^r + 1 \\
 S_r &= 6n_r + 1 = 6 \cdot 9^r + 1 \\
 C_r &= 3n_r^2 + 3n_r + 1 = 3 \cdot 81^r + 3 \cdot 9^r + 1.
 \end{aligned}$$

Invariant	Volume 729^r	Face 81^r	Edge 9^r	Point 1
Q_r	8	12	6	1
B_r	0	24	0	2
I_r	3	-3	0	0
$E_{\text{ext},r}$	0	6	0	0
$\text{beta}_{1,r}$	2	-3	0	1
S_r	0	0	6	1
C_r	0	3	3	1

The coefficient table is a spectral fingerprint. Any measured or computed sequence that belongs to this exact cubical recursion must lie in the span of 729^r , 81^r , 9^r , and 1. The channel coefficients identify which geometric dimensions contribute to that

observable.

8. The Seven-Sheet Cell3 Kernel

Define $U_n(z) = 1 + z + \dots + z^{(n-1)}$. The coefficient of z^s in $U_n(z)^3$ counts the number of three-coordinate addresses whose component sum is s . For the three-position Cell3 address set $\{-1, 0, 1\}$, an index shift gives the same coefficients as $U_3(z)^3$.

$$\begin{aligned}\text{Lambda}_3(z) &= U_3(z)^3 = (1 + z + z^2)^3 \\ &= 1 + 3z + 6z^2 + 7z^3 + 6z^4 + 3z^5 + z^6.\end{aligned}$$

The seven coefficients 1, 3, 6, 7, 6, 3, 1 are the exact populations of the seven laminar sheets. Their sum is 27, the number of local states. The central sheet contains seven states, and the coefficients are symmetric because address inversion maps grade λ to $-\lambda$.

The elementary geometric-series identity

$$U_9(z) = U_3(z) \ U_3(z^3)$$

implies the three-dimensional factorization

$$\text{Lambda}_9(z) = \text{Lambda}_3(z) \ \text{Lambda}_3(z^3).$$

A nine-position axis is therefore built from one seven-sheet spectrum at unit grade and a second copy dilated by a factor of 3. Repeating the identity gives the exact recursive hierarchy

$$\text{Lambda}_{\{9^r\}}(z) = \text{product from } j=0 \text{ to } 2r-1 \text{ of } \text{Lambda}_3(z^{(3^j)}).$$

Two ternary stages per closure. Since $9=3^2$, every length multiplication by 9 inserts two consecutive ternary laminar factors. The seven-sheet Cell3 complex is the irreducible spectral kernel of the complete radix-9 hierarchy.

Exact laminar spectra from $\Lambda_n(z)=U_n(z)^3$

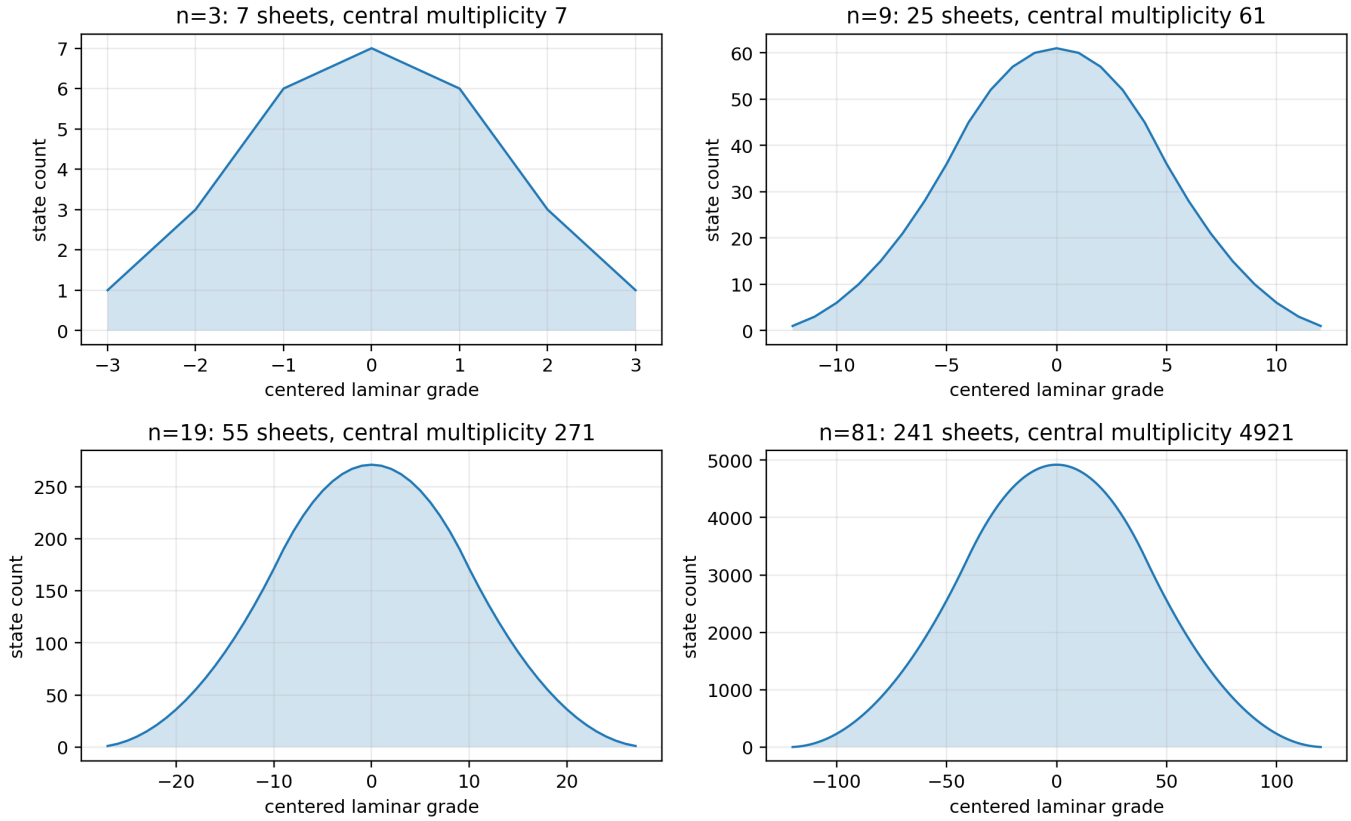


Figure 7. Exact laminar coefficient spectra for $n=3, 9, 19$, and 81 positions per axis. The $n=3$ panel is the seven-sheet Cell3 kernel; $n=9$ is the Supergroup-center spectrum; $n=19$ is the complete 6859-state refined Supergroup; $n=81$ is the second-order coarse closure spectrum.

9. Fourier Harmonics and Subharmonics

The Fourier transform of U_n is a phase factor times the Dirichlet kernel. The phase factor changes only complex phase, so the amplitude magnitude is

$$D_n(\omega) = \sin(n\omega/2) / \sin(\omega/2).$$

Because $\Lambda_n = U_n^3$, the laminar amplitude and power spectra are

$$\begin{aligned} \Lambda_{\hat{n}}(\omega) &= D_n(\omega)^3 \\ P_n(\omega) &= |D_n(\omega)|^6. \end{aligned}$$

The zeros occur at $\omega = 2\pi k/n$ for integer k not divisible by n . Increasing n from 9^r to 9^{r+1} inserts eight new zero intervals between every inherited pair. The old zeros remain because every old index is also a new index multiplied by 9.

The exact factorization $D_9(\omega) = D_3(\omega)D_3(3\omega)$ separates the two ternary stages inside one Lucas closure. More generally

$$D_{\{9^r\}}(\omega) = \text{product from } j=0 \text{ to } r-1 \text{ of } D_9(9^j \omega).$$

Normalized sixth-power Dirichlet spectra: harmonic inheritance and subharmonic birth

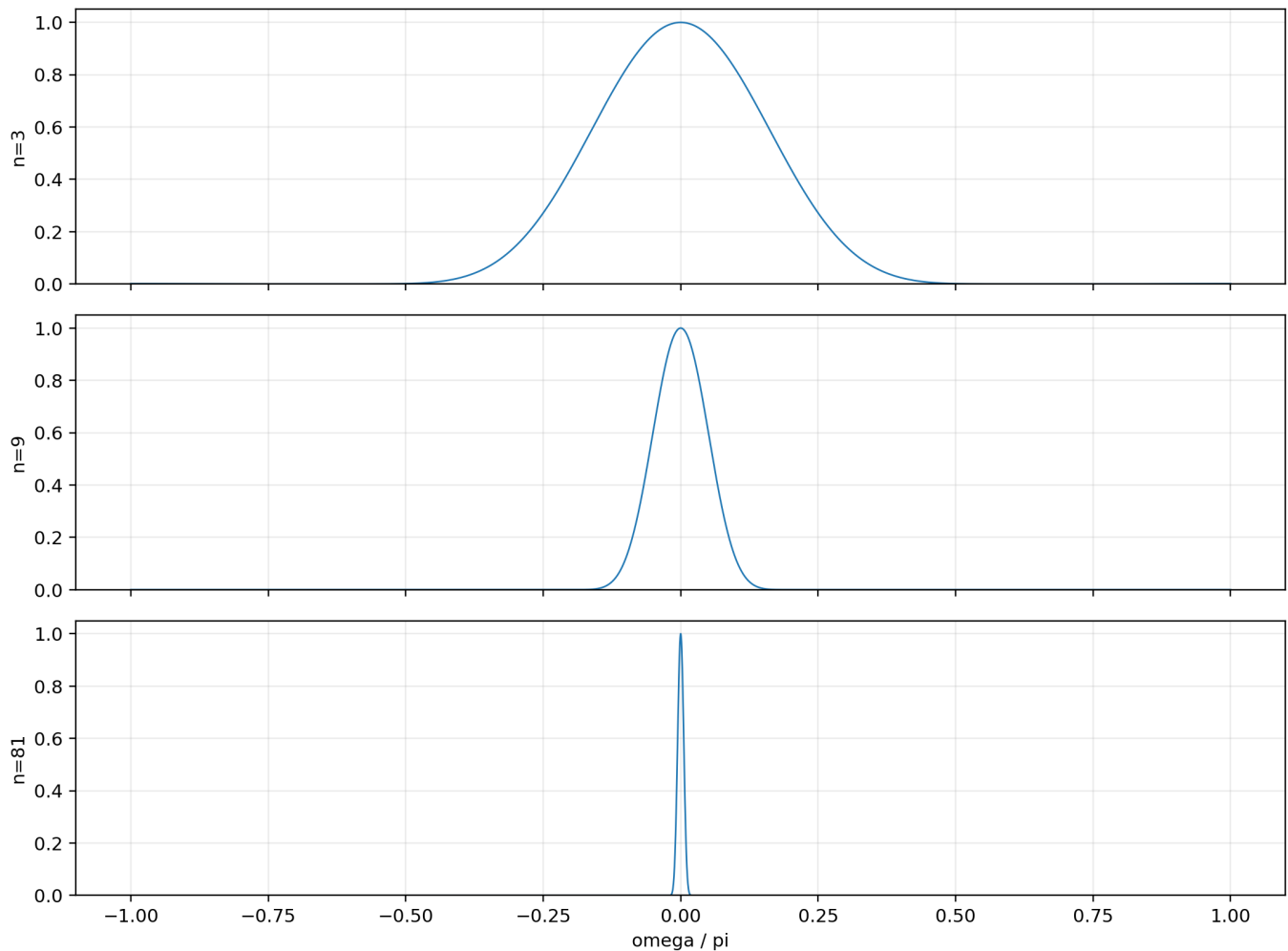


Figure 8. Numerically evaluated normalized power spectra $|D_n|^6/n^6$ for $n=3$, 9, and 81. Every increase in closure depth preserves inherited peaks and inserts a denser family of subharmonic bands and exact zeros.

The spatial classes can be read as inherited harmonics, intermediate one-third harmonics, and closure-native one-ninth subharmonics. These names refer to the residue structure generated inside one radix-9 step; the complete hierarchy repeats the same classification at every scale.

10. Exact 729-Band Folding

When a lower-level object is assembled nine times along each axis, the child mode index can be decomposed into a parent index and a radix-9 residue. The residue is a three-component vector ρ in $(\mathbb{Z}/9\mathbb{Z})^3$. There are $9^3=729$ residue sectors per parent mode.

$$729 = 1 + 26 + 702.$$

The three sectors are defined exactly as follows:

Inherited sector. $\rho=(0,0,0)$. Multiplying the parent mode index by 9 reproduces its wavevector exactly at the child scale.

Lucas-intermediate sectors. ρ belongs to $\{0,3,6\}^3$ but is not zero. These 26 sectors form a complete noncentral $3 \times 3 \times 3$ ternary scaffold embedded in the radix-9 cube.

Closure-native sectors. At least one residue component is not a multiple of 3. The remaining 702 sectors exist only after the full radix-9 closure is resolved.

Exact 729-sector folding: 1 inherited + 26 intermediate + 702 native

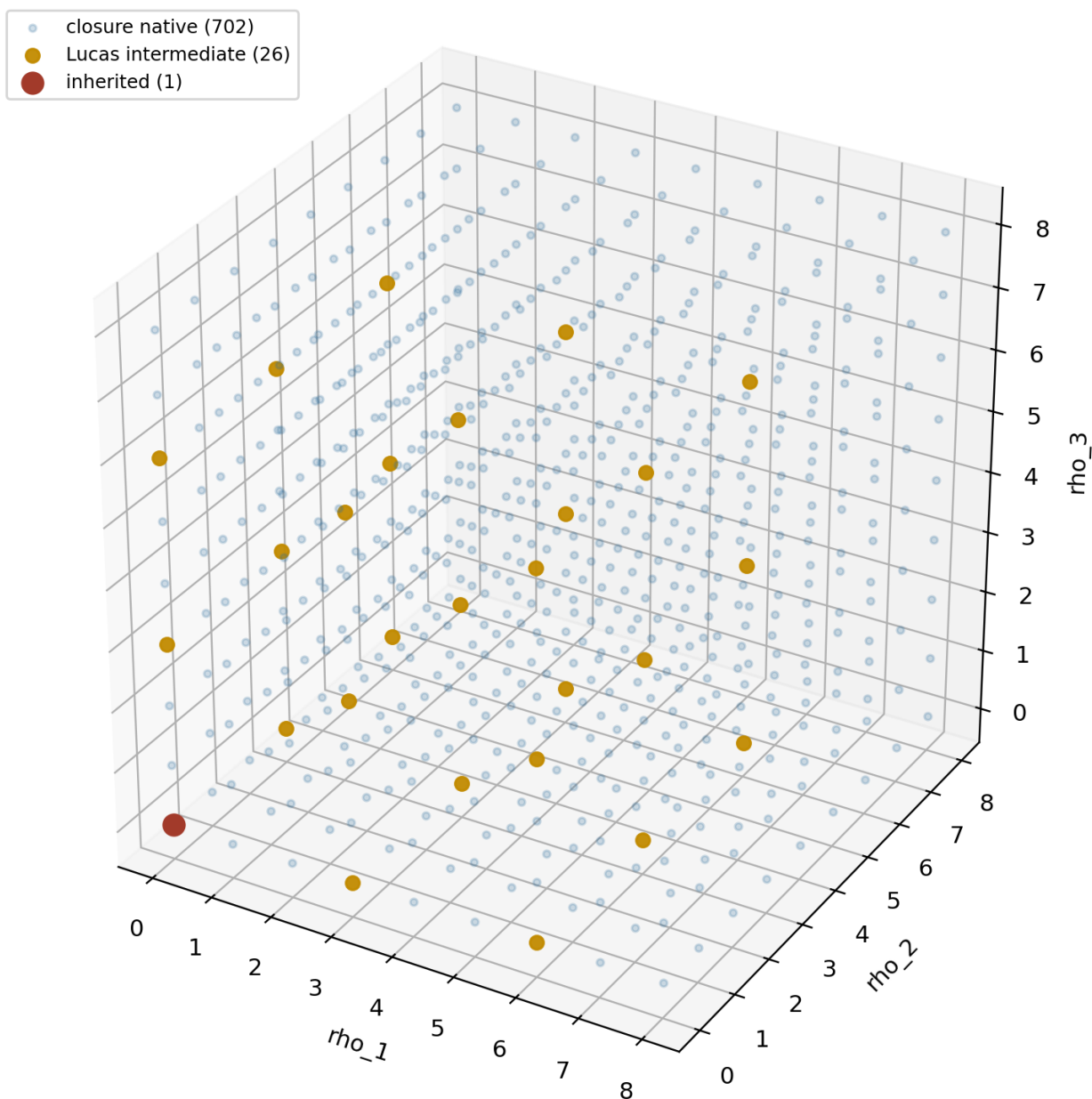


Figure 9. The complete radix-9 residue cube. The central inherited sector is red, the 26 Lucas-intermediate sectors are gold, and the 702 closure-native sectors are blue.

Sector	Count	chi=+1	chi=-1	Meaning
Inherited	1	1	0	exact parent mode
Lucas-intermediate	26	13	13	noncentral ternary scaffold
Closure-native	702	351	351	new radix-9 sectors
Total	729	365	364	one central polarity excess

Using $\text{chi}=(-1)^{(\text{rho}_1+\text{rho}_2+\text{rho}_3)}$, every noncentral class is exactly polarity balanced. The only uncompensated state is the inherited central mode. This is the spectral counterpart of the one-state polarity excess of every odd cubical lattice.

11. Reciprocal-Space Mode Nesting

The reciprocal metric determines the physical splitting of integer mode addresses. For absolute closure order r , the allowed periodic wavevectors are

$$k_r(n) = (2\pi/9^r) B(-T) n, \quad n \in \mathbb{Z}^3.$$

Their squared magnitudes are

$$kappa_r^2(n) = ((2\pi)^2/81^r) n^T R n.$$

Two exact scale identities follow immediately:

$$\begin{aligned} kappa_{r+1}^2(n) &= kappa_r^2(n)/81 \\ k_{r+1}(9n) &= k_r(n), \text{ and therefore } kappa_{r+1}^2(9n) = kappa_r^2(n). \end{aligned}$$

The first identity contracts the entire reciprocal spectrum by $1/81$. The second embeds every parent mode exactly into the child spectrum. All mode indices not divisible by 9 are new subharmonic states. Since $q(-n)=q(n)$, antipodal modes remain exactly paired.

The unequal Cell3 metric encoded in R splits mode addresses that would be degenerate in an orthogonal cube. The recursion creates the band families; the oblique Mass-Bridge geometry supplies their direction-dependent splitting. The normalized ordering of $q(n)$ is unchanged by recursive scale because every q value receives the same factor $1/81$ per closure.

Lowest antipodal-quotiented reciprocal modes at Lucas-Supergroup order

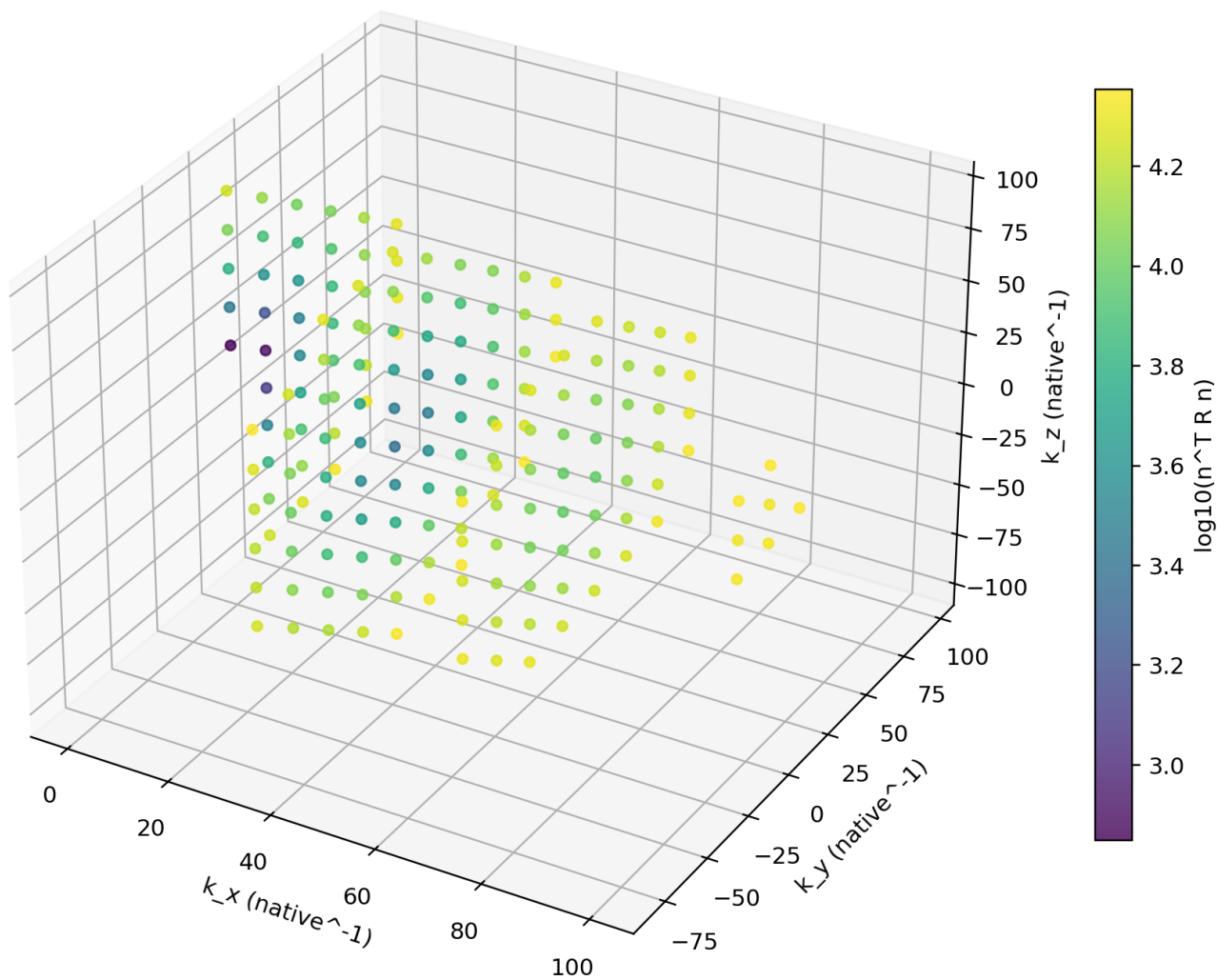


Figure 10. Lowest reciprocal modes at order L_1 , quotiented by antipodal inversion. Coordinates are computed directly from $(2\pi/9)B^{(-T)}n$; color represents $\log_{10}(n^{\wedge T} R n)$.

rank	n	$q=n^{\wedge T} R n$	κ_1^2	lambda	chi
1	[0, 0, 1]	705.188414265	343.700280347	1	-1
2	[0, 1, 1]	775.689190823	378.061503784	2	1
3	[0, 1, 0]	1266.057428561	617.061035504	1	-1
4	[0, 1, 2]	1695.697781614	826.462532757	3	-1
5	[1, 0, 0]	2377.512203001	1158.770612463	1	-1
6	[1, 0, 1]	2749.525516322	1340.085389471	2	1
7	[0, 0, 2]	2820.753657058	1374.801121387	2	1
8	[1, 1, 1]	2820.818862895	1374.832901921	3	-1
9	[0, 2, 2]	3102.756763292	1512.246015136	4	1
10	[0, 1, -1]	3166.802494829	1543.461127918	0	1
11	[0, 2, 1]	3378.304824504	1646.544798230	3	-1
12	[1, 1, 2]	3407.652352741	1660.848427556	4	1
13	[1, 0, -1]	3415.875718210	1664.856396149	0	1

rank	n	q=n^T R n	kappa_1^2	lambda	chi
14	[1, -1, -1]	3485.583924754	1698.831330574	-1	-1
15	[1, -1, 0]	3642.777061548	1775.445358955	0	1
16	[1, 1, 0]	3644.362201577	1776.217936980	2	1
17	[0, 1, 3]	4026.083200933	1962.264122424	4	1
18	[0, 2, 3]	4237.585530609	2065.347792735	5	-1
19	[1, 0, 2]	4531.915658171	2208.800727173	3	-1
20	[1, -1, -2]	4738.767616488	2309.617862886	-2	1
21	[1, 2, 2]	4815.503904434	2347.018198947	5	-1
22	[0, 2, 0]	5064.229714245	2468.244142017	2	1
23	[1, -1, 1]	5210.347026872	2539.459948030	1	-1
24	[1, 1, 3]	5404.862671117	2634.264513884	5	-1

12. Unique Recursive Factorization of Arbitrary Volumes

Every positive axis count has a unique radix-9 factorization

$$a=9^{(r_a)} a_{\text{bar}}, b=9^{(r_b)} b_{\text{bar}}, c=9^{(r_c)} c_{\text{bar}},$$

where none of a_{bar} , b_{bar} , or c_{bar} is divisible by 9. The maximum common exact closure depth is

$$r_{\text{common}}=\min(r_a, r_b, r_c).$$

The one-dimensional generating function factors as

$$U_a(z)=[\text{product from } j=0 \text{ to } r_a-1 \text{ of } U_9(z^{(9^j)})] U_{\{a_{\text{bar}}\}}(z^{(9^{r_a})}),$$

with corresponding expressions for b and c . Thus every aligned rectangular volume has a unique decomposition into exact radix-9 closure factors multiplied by a nonrecursive outer envelope. Equal axis depths give isotropic closure. Unequal depths give anisotropic closure and directional subharmonic depth.

Dimensions	Depths (r_a, r_b, r_c)	Residual envelope	Common depth	Morphology
1 x 1 x 1	(0, 0, 0)	(1, 1, 1)	0	isotropic exact closure
2 x 3 x 5	(0, 0, 0)	(2, 3, 5)	0	nonrecursive aligned envelope
9 x 9 x 9	(1, 1, 1)	(1, 1, 1)	1	isotropic exact closure
18 x 45 x 72	(1, 1, 1)	(2, 5, 8)	1	common closure plus anisotropic envelope
81 x 27 x 9	(2, 1, 1)	(1, 3, 1)	1	common closure plus anisotropic envelope
81 x 81 x 81	(2, 2, 2)	(1, 1, 1)	2	isotropic exact closure
729 x 81 x 9	(3, 2, 1)	(1, 1, 1)	1	common closure plus anisotropic envelope

Exact and anisotropic recursive envelopes in the authoritative oblique basis

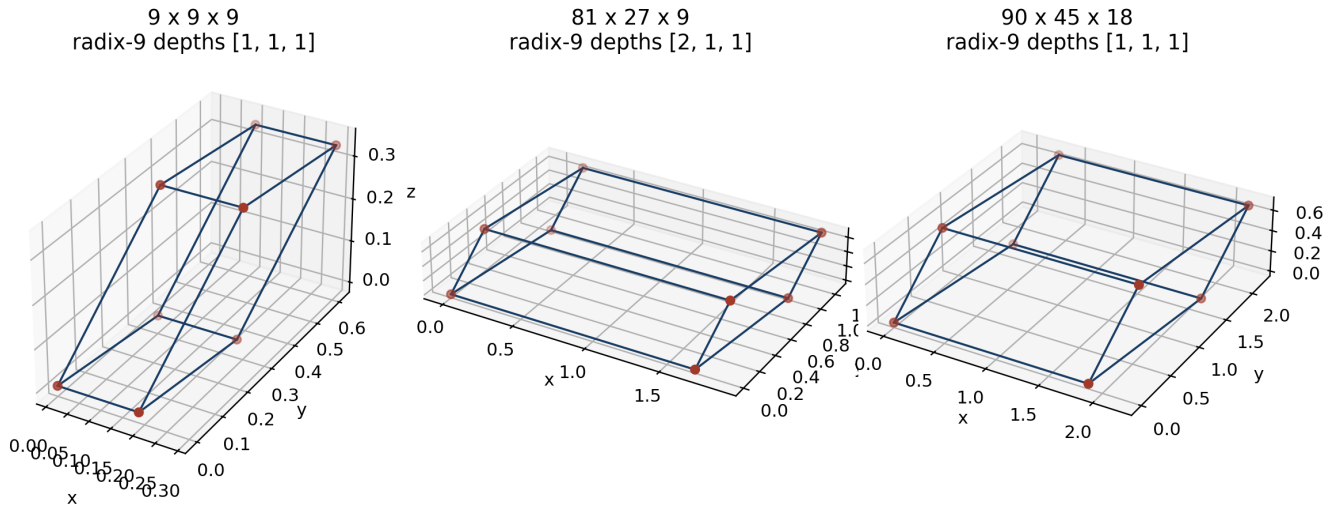


Figure 11. Three exact oblique envelopes generated from B. The 9 x 9 x 9 object closes isotropically. The 81 x 27 x 9 object has radix-9 depths (2,1,1). The 90 x 45 x 18 object has one common closure factor and a residual 10 x 5 x 2 envelope.

Different axis depths create a scale-dependent dimensional crossover. Before the shortest axis closes, the mode family behaves as a rod-like or sheet-like spectrum. Once all three axes share a radix-9 factor, the complete volumetric 729-sector folding appears. This transition is exact and determined by integer factorization.

13. Recursive Energy-Band Structure

The established cubic aggregate law can be written in normalization-independent form as

$$E(m) = C m^2 (3m-1) = 3C m^3 - C m^2.$$

Here m is the linear Lucas-Supergroup multiple and C is the chosen energy normalization. With $C = \epsilon_{LS}$, $E/\epsilon_{LS} = m^2(3m-1)$. Since $\epsilon_{LS} = 729 \epsilon_{Cell3}$, the same values in Cell3 energy units are $729m^2(3m-1)$.

On the exact recursive subsequence $m = 9^r$,

$$E_r = 3C \cdot 729^r - C \cdot 81^r.$$

The energy has two exact branches: a volumetric bulk branch proportional to 729^r and a surface-deficit branch proportional to 81^r . Their ratio is

$$\text{surface deficit} / \text{bulk} = 1/(3 \cdot 9^r).$$

The relative correction decreases by exactly 1/9 per closure. Because the sequence contains only two exponential branches, it obeys the second-order recurrence

$$E_{\{r+2\}} = 810 E_{\{r+1\}} - 59049 E_r.$$

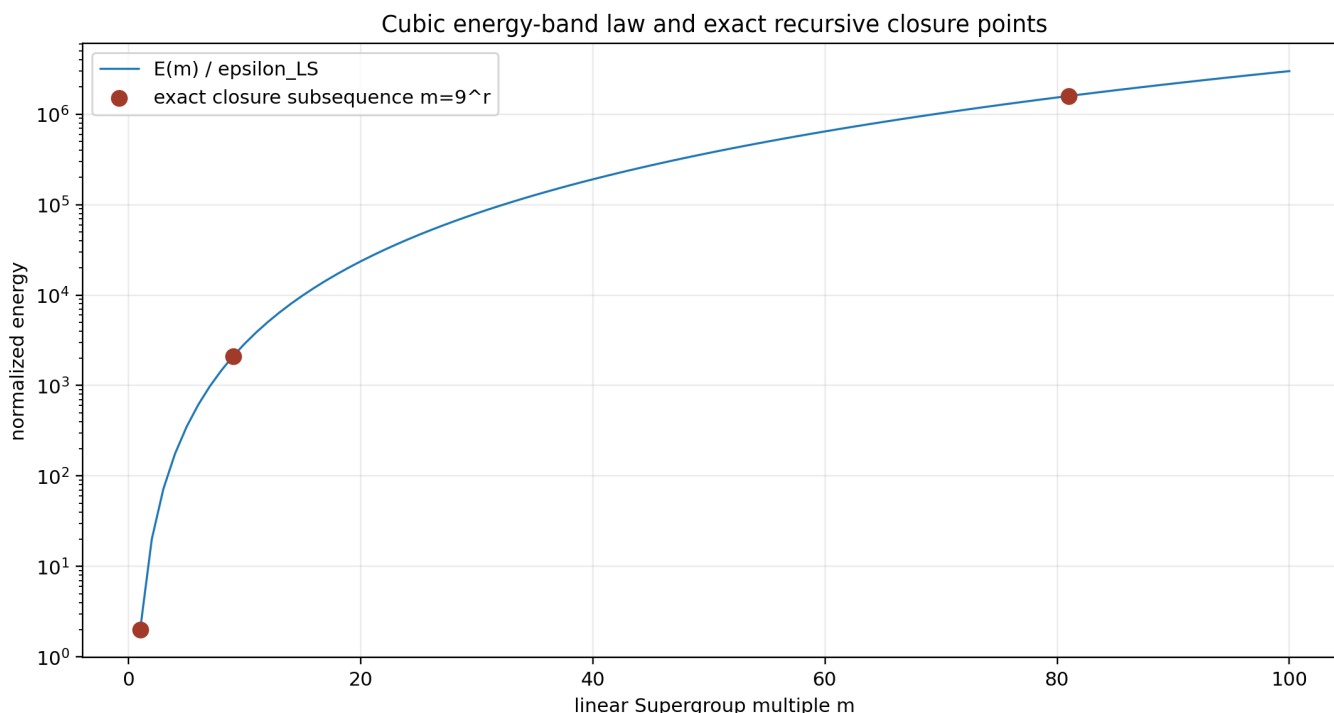


Figure 12. Exact normalized energy law $E/\epsilon_{LS}=m^2(3m-1)$ for $m=1$ through 100. Red points mark the recursive closure subsequence $m=9^r$ within the plotted range.

r	$m=9^r$	bulk $3 \cdot 729^r$	surface 81^r	E/ϵ_{LS}	E/ϵ_{Cell3}
0	1	3	1	2	1458
1	9	2187	81	2106	1535274
2	81	1594323	6561	1587762	1157478498
3	729	1162261467	531441	1161730026	846901188954
4	6561	847288609443	43046721	847245562722	617642015224338

The recursive energy ladder is not uniformly spaced. Its spacing grows cubically with linear size, while its finite-volume deficit remains a separate quadratic branch. The branch decomposition is the energy analogue of the morphological volume and face channels.

14. Universal Four-Channel Recurrence

Any local cubical observable that is a linear combination of volume, face, edge, and point contributions has the recursive form

$$X_r = A \cdot 729^r + B \cdot 81^r + C \cdot 9^r + D.$$

The characteristic polynomial is

$$(\lambda - 729)(\lambda - 81)(\lambda - 9)(\lambda - 1) \\ = \lambda^4 - 820\lambda^3 + 67158\lambda^2 - 597780\lambda + 531441.$$

Therefore every such sequence satisfies

$$X_{r+4} = 820X_{r+3} - 67158X_{r+2} + 597780X_{r+1} - 531441X_r.$$

This recurrence is a direct diagnostic for exact Lucas recursion. Four consecutive scale channels determine all later values. Failure of a computed sequence to satisfy the recurrence identifies either a nonlocal contribution, a change of boundary condition, an anisotropic scale step, or a departure from aligned cubical filling.

15. Cycle Space and Path-Entropy Spectrum

The graph formed by joining centers of face-adjacent cells has $V=m^3$ vertices and $E=3m^2(m-1)$ edges. Since the graph is connected, its first cycle rank is $E-V+1$:

$$\text{beta}_1(m)=2m^3-3m^2+1.$$

On the exact recursive subsequence $m=9^r$,

$$\text{beta}_1, r=2 \cdot 729^r - 3 \cdot 81^r + 1.$$

The cycle space therefore contains a volumetric route branch, a boundary-suppression branch, and one invariant closure mode. Its density approaches two independent cycles per occupied coarse cell:

$$\text{beta}_1/m^3 = 2 - 3/m + 1/m^3 \rightarrow 2.$$

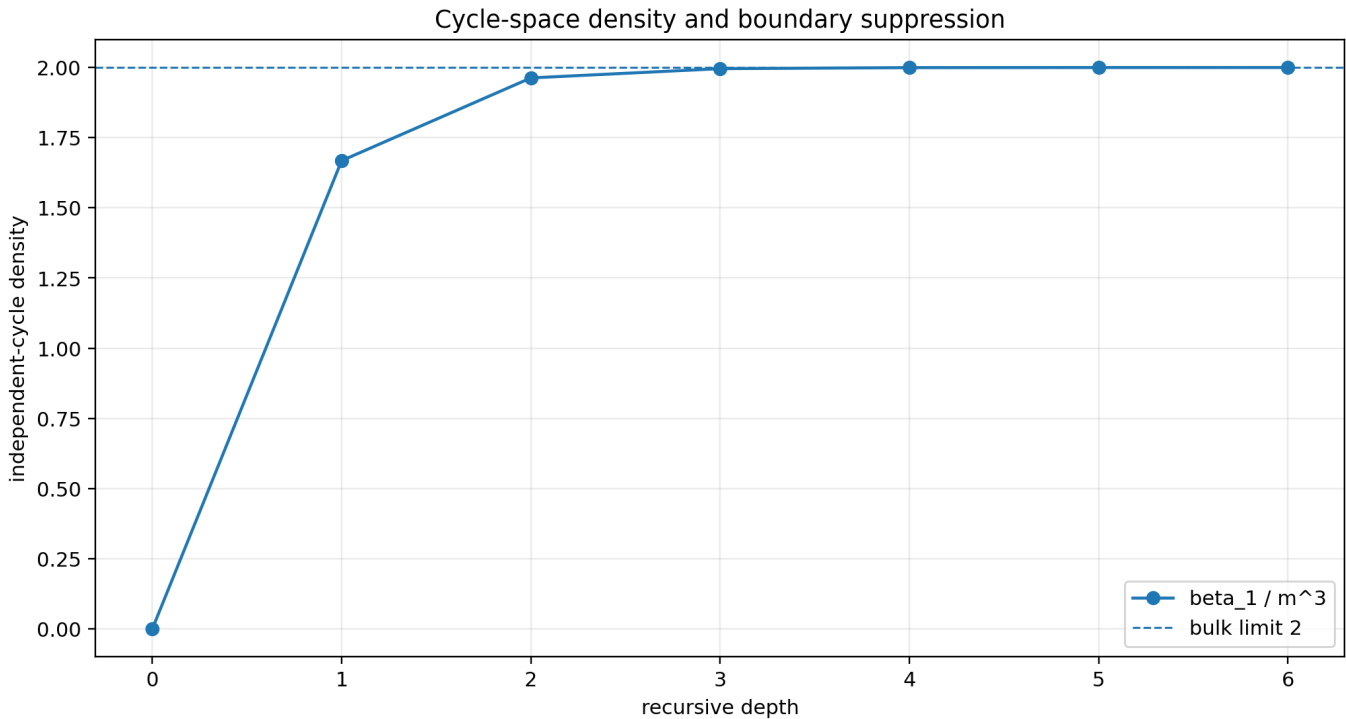


Figure 13. Independent-cycle density along exact recursive scales. The density converges rapidly to the bulk limit 2 as the surface correction contracts by 1/9 per closure.

A machine-independent path-capacity measure is beta_1 itself. If each independent cycle is assigned two equal binary states, the corresponding combinatorial entropy is $H=\text{beta}_1 \ln 2$ nats. This binary encoding is a clearly stated counting convention; the exact geometric result is the cycle rank and its 729/81/1 channel decomposition.

16. New Recursive Spectral Properties

The harmonic and subharmonic analysis exposes properties that are not visible from the raw census alone. Each property below follows directly from the exact radix-9 geometry and the locked oblique metric.

16.1 Exact discrete scale invariance

The hierarchy is invariant under simultaneous length multiplication by 9 and resolution relabeling by one Lucas level. This is stronger than geometric similarity: the complete cell generating polynomial, refined state law, boundary/interface relation, laminar factorization, and mode embedding are preserved in their corresponding coarse variables.

16.2 Two-stage ternary spectral birth

One closure factor of 9 equals two factors of 3. The identity $\Lambda_9 = \Lambda_3(z)\Lambda_3(z^3)$ proves that each closure creates one new ternary family at the immediate scale and a second family at the three-times-dilated scale. After r closures there are exactly $2r$ nested ternary factors.

16.3 Exact mode inheritance and spectral-gap contraction

Every parent wavevector is copied exactly into the child spectrum by index multiplication $n \rightarrow 9n$. The entire squared-wavevector spectrum contracts by $1/81$. Consequently every reciprocal spectral gap also contracts by $1/81$ before the newly born subharmonic states are inserted between inherited levels.

16.4 Fixed band-birth census

Each parent mode generates 729 residue sectors. One is inherited and 728 are new at the child indexing resolution. The new set divides exactly into 26 Lucas-intermediate modes and 702 closure-native modes. This birth census repeats independently around every parent mode and at every recursive scale.

16.5 Noncentral polarity cancellation

The 26 intermediate sectors divide $13+13$ by polarity, and the 702 native sectors divide $351+351$. Any complete noncentral spectral shell therefore has zero net checkerboard polarity before metric weighting. The inherited center supplies the unique one-state excess.

16.6 Scale-invariant anisotropic splitting

The reciprocal quadratic form $q(n) = n^T R n$ determines directional splitting. Since recursive closure multiplies every q value by the same factor $1/81$, all normalized anisotropic ratios are invariant across scale. A mode ordering measured at one closure level is reproduced at every higher level, with additional subharmonic states inserted between inherited modes.

16.7 Boundary-renormalization law

Every surface-to-volume correction contracts by exactly $1/9$ per complete closure. Edge-to-volume and point-to-volume corrections contract by $1/81$ and $1/729$. Finite-volume observables therefore approach their bulk limits through a known hierarchy of exact correction exponents rather than an undifferentiated convergence.

16.8 Dimensional crossover in anisotropic arrays

When axis depths differ, the spectrum closes first along the most highly divisible axes. One-axis recursion creates line-like band multiplication; two-axis recursion creates sheet-like multiplication; equal recursion on all three axes creates the full 729-sector volumetric band. The common radix-9 valuation determines the completed three-dimensional closure depth.

16.9 Finite-difference and recurrence spectroscopy

The energy law has constant third finite difference in linear size m , while scale-indexed recursive observables satisfy fixed exponential recurrences. These two diagnostics separate ordinary polynomial aggregation from exact discrete-scale recursion. Applying both tests to a measured sequence identifies whether its organizing variable is linear size, recursive depth, or a mixture of the two.

16.10 Closure selection rules

A perturbation that preserves the parent index modulo 9 remains in the inherited sector. Residues that are multiples of 3 but not all zero occupy the Lucas-intermediate scaffold. All other residues are closure-native. This residue rule supplies an exact address-space selection law for coupling calculations, mode classification, and band-resolved experiments.

17. Computational and Experimental Examination Protocol

The recursive spectral structure can be examined without fitting free geometric parameters. The required inputs are the locked Cell3 basis B , an aligned integer array size, and the chosen boundary condition.

17.1 Geometry and census test

Construct the array directly from integer addresses. Count cells by dimension, refined states, internal interfaces, exposed faces, and center-graph cycles. At exact isotropic closure the coarse census must reproduce (1000,2700,2430,729), and the boundary/interface identity must reproduce $1946=1944+2$ at the corresponding nine-cell resolution.

17.2 Laminar polynomial test

Compute the histogram of coordinate sums. Compare it exactly with the coefficients of $U_n(z)^3$. For $n=9^r$, factor the polynomial into the $2r$ seven-sheet kernels $\Lambda_3(z^{(3^j)})$. Equality of every coefficient verifies the complete ternary recursion.

17.3 Fourier band test

Evaluate the discrete Fourier transform of the laminar sequence. Confirm zeros at $2\pi k/n$, parent-peak inheritance, and the insertion of eight child intervals per parent interval at each radix-9 step. Normalize by n^6 to compare shapes across scale.

17.4 Reciprocal metric test

Enumerate integer mode addresses n , compute $q=n^T R n$, and quotient antipodal pairs. At successive closure levels verify $k_{r+1}(9n)=k_r(n)$ and $\kappa_{r+1}^2(n)=\kappa_r^2(n)/81$. The normalized anisotropic q ratios must remain unchanged.

17.5 Residue-sector test

Reduce child mode indices modulo 9. Confirm the exact populations 1, 26, and 702 and the polarity divisions 1/0, 13/13, and 351/351. This test is purely arithmetic and is independent of the metric.

17.6 Energy and route-space test

For linear multiples m , verify the cubic law $E/C=m^2(3m-1)$ and its constant third finite difference. On the recursive subsequence $m=9^r$, verify the second-order recurrence. Independently compute $\beta_1=2m^3-3m^2+1$ and its convergence to two cycles per cell.

18. Integrated Interpretation

The Lucas hierarchy combines two exact structures. The first is the morphological spectrum $\{729,81,9,1\}$, which separates bulk, surface, edge, and point content. The second is the 729-sector folding census $1+26+702$, which separates inherited, Lucas-intermediate, and closure-native spatial modes.

The seven-sheet Cell3 polynomial is the common kernel joining these structures. Its direct coefficients describe laminar state populations. Its Fourier transform describes harmonic amplitudes. Its radix-3 dilations build the radix-9 closure. Its checkerboard parity produces exact noncentral cancellation. The direct metric determines the cell geometry, while the reciprocal metric converts integer mode addresses into anisotropically split spectral values.

Central spectral theorem. Exact Lucas recursion multiplies length by 9, decomposes morphology into eigenchannels $\{729,81,9,1\}$, inserts two ternary harmonic generations, folds every parent mode into 729 child sectors with census $1+26+702$, contracts reciprocal squared magnitudes by $1/81$, and preserves every parent mode through the exact embedding $n \rightarrow 9n$.

Arbitrary aligned volumes do not fall outside the theory. Their axis counts factor uniquely into radix-9 closure depths and a residual envelope. The common depth determines full volumetric recursion; unequal depths determine anisotropic band splitting and dimensional crossover. The recursive spectrum is therefore defined for every aligned rectangular volume, not only for perfect cubes.

The resulting framework supplies a complete mathematical basis for scale-recursive mode classification, harmonic and subharmonic band prediction, finite-volume correction analysis, energy-ladder organization, and path-space capacity across the K-Field Lucas hierarchy.

Appendix A. Formula and Theorem Ledger

ID	Formula	Interpretation
A1	$B_{out}=9B \text{ diag}(a,b,c)$	outer basis of aligned SG array
A2	$F_{\{a,b,c\}}(t)=[(a+1)+at][(b+1)+bt][(c+1)+ct]$	coarse cellular generating polynomial
A3	$Q_{SG}=(2a+1)(2b+1)(2c+1)$	coarse refined-state count
A4	$B_{SG}=8(ab+ac+bc)+2$	coarse boundary states
A5	$I_{SG}=3abc-(ab+ac+bc)$	internal SG contacts
A6	$\beta_1=2abc-(ab+ac+bc)+1$	coarse center-graph cycle rank
A7	$n_r=9^r$	absolute recursive linear scale
A8	$F_r=729^r \text{ abc}(1+t)^3+81^r \Sigma_2(1+t)^2+9^r \Sigma_1(1+t)+1$	morphological eigendecomposition
A9	$\Lambda_3=(1+z+z^2)^3$	seven-sheet kernel
A10	$\Lambda_{\{9^r\}}=\text{product } \Lambda_3(z^{(3^j)}), j=0..2r-1$	recursive laminar factorization
A11	$D_n=\sin(n \omega/2)/\sin(\omega/2)$	Dirichlet kernel
A12	$P_n= D_n ^6$	laminar power spectrum
A13	$729=1+26+702$	radix-9 folding census
A14	$k_r=(2\pi/9^r)B^{(-T)}n$	reciprocal mode vector
A15	$\kappa_r^2=(2\pi)^2 81^{(-r)}n^T R n$	reciprocal eigenvalue
A16	$k_{\{r+1\}}(9n)=k_r(n)$	exact parent-mode inheritance
A17	$E(m)=C m^2(3m-1)$	cubic energy-band law
A18	$E_r=3C729^r-C81^r$	recursive bulk/surface split
A19	$\beta_1=2m^3-3m^2+1$	cycle-rank law
A20	$X_r=A729^r+B81^r+C9^r+D$	universal local recursive observable

Appendix B. Complete 27-Node Cell3 Ledger

Every local midpoint state is listed below. Positions are computed as $p=(1/2)B$ s in native units. The address order is (s_a,s_b,s_c).

address	class	lambda	chi	x	y	z
-1-1-1	vertex	-3	-1	-0.016418079602	-0.033526107714	-0.018828573589
-1-1+0	edge midpoint	-2	1	-0.012326937410	-0.018144916524	0.000000000000
-1-1+1	vertex	-1	-1	-0.008235795218	-0.002763725333	0.018828573589
-1+0-1	edge midpoint	-2	1	-0.014489671542	-0.015381191190	-0.018828573589
-1+0+0	face center	-1	-1	-0.010398529350	0.000000000000	0.000000000000
-1+0+1	edge midpoint	0	1	-0.006307387158	0.015381191190	0.018828573589
-1+1-1	vertex	-1	-1	-0.012561263482	0.002763725333	-0.018828573589
-1+1+0	edge midpoint	0	1	-0.008470121290	0.018144916524	0.000000000000
-1+1+1	vertex	1	-1	-0.004378979098	0.033526107714	0.018828573589
+0-1-1	edge midpoint	-2	1	-0.006019550252	-0.033526107714	-0.018828573589
+0-1+0	face center	-1	-1	-0.001928408060	-0.018144916524	0.000000000000
+0-1+1	edge midpoint	0	1	0.002162734132	-0.002763725333	0.018828573589
+0+0-1	face center	-1	-1	-0.004091142192	-0.015381191190	-0.018828573589
+0+0+0	center	0	1	0.000000000000	0.000000000000	0.000000000000
+0+0+1	face center	1	-1	0.004091142192	0.015381191190	0.018828573589
+0+1-1	edge midpoint	0	1	-0.002162734132	0.002763725333	-0.018828573589
+0+1+0	face center	1	-1	0.001928408060	0.018144916524	0.000000000000
+0+1+1	edge midpoint	2	1	0.006019550252	0.033526107714	0.018828573589
+1-1-1	vertex	-1	-1	0.004378979098	-0.033526107714	-0.018828573589
+1-1+0	edge midpoint	0	1	0.008470121290	-0.018144916524	0.000000000000
+1-1+1	vertex	1	-1	0.012561263482	-0.002763725333	0.018828573589
+1+0-1	edge midpoint	0	1	0.006307387158	-0.015381191190	-0.018828573589
+1+0+0	face center	1	-1	0.010398529350	0.000000000000	0.000000000000
+1+0+1	edge midpoint	2	1	0.014489671542	0.015381191190	0.018828573589
+1+1-1	vertex	1	-1	0.008235795218	0.002763725333	-0.018828573589
+1+1+0	edge midpoint	2	1	0.012326937410	0.018144916524	0.000000000000
+1+1+1	vertex	3	-1	0.016418079602	0.033526107714	0.018828573589

Appendix C. Recursive Closure Ledger

C.1 Cellular census by recursive order

r	$n=9^r$	f0 vertices	f1 edges	f2 faces	f3 volumes
1	9	1,000	2,700	2,430	729
2	81	551,368	1,633,932	1,614,006	531,441
3	729	389,017,000	1,165,452,300	1,163,855,790	387,420,489
4	6,561	282,558,696,328	847,546,909,452	847,417,749,606	282,429,536,481
5	59,049	205,901,592,625,000	617,694,317,167,500	617,683,856,637,150	205,891,132,094,649
6	531,441	150,095,482,587,202,888	450,285,600,469,810,572	450,284,753,179,606,806	150,094,635,296,999,121

C.2 Refined-state, boundary, and interface census

r	Q refined states	boundary states	internal interfaces	exposed faces
1	6,859	1,946	1,944	486
2	4,330,747	157,466	1,574,640	39,366
3	3,105,745,579	12,754,586	1,160,667,144	3,188,646
4	2,259,952,891,867	1,033,121,306	847,159,469,280	258,280,326
5	1,647,170,898,524,299	83,682,825,626	617,662,935,930,744	20,920,706,406
6	1,200,760,471,533,619,387	6,778,308,875,546	450,283,058,602,387,920	1,694,577,218,886

C.3 Recursive hierarchy and laminar-cycle census

r	Lucas SG	Lucas Groups	Cell3 volumes	beta_1	sheets	central sheet
1	1	27	729	1,216	55	271
2	729	19,683	531,441	1,043,200	487	19,927
3	531,441	14,348,907	387,420,489	773,246,656	4,375	1,596,511
4	387,420,489	10,460,353,203	282,429,536,481	564,729,932,800	39,367	129,159,847
5	282,429,536,481	7,625,597,484,987	205,891,132,094,649	411,771,803,836,096	354,295	10,460,530,351
6	205,891,132,094,649	5,559,060,566,555,523	150,094,635,296,999,121	300,188,423,305,388,800	3,188,647	847,290,203,767

Appendix D. Lowest Reciprocal Modes

The table is antipodal-quotiented: each pair n and $-n$ appears once. q is in native inverse-length squared units. κ_1^2 includes the order- L_1 factor $(2\pi/9)^2$.

rank	n	q	k _x	k _y	k _z	κ_1^2	chi
1	[0, 0, 1]	705.18841426	0.0000000	0.0000000	18.5391553	343.70028035	-1
2	[0, 1, 1]	775.68919082	0.0000000	19.2376664	2.8237734	378.06150378	1
3	[0, 1, 0]	1266.05742856	0.0000000	19.2376664	-15.7153819	617.06103550	-1
4	[0, 1, 2]	1695.69778161	0.0000000	19.2376664	21.3629288	826.46253276	-1
5	[1, 0, 0]	2377.51220300	33.5687710	-3.5676267	-4.3795281	1158.77061246	-1
6	[1, 0, 1]	2749.52551632	33.5687710	-3.5676267	14.1596272	1340.08538947	1
7	[0, 0, 2]	2820.75365706	0.0000000	0.0000000	37.0783107	1374.80112139	1
8	[1, 1, 1]	2820.81886289	33.5687710	15.6700397	-1.5557547	1374.83290192	-1
9	[0, 2, 2]	3102.75676329	0.0000000	38.4753327	5.6475469	1512.24601514	1
10	[0, 1, -1]	3166.80249483	0.0000000	19.2376664	-34.2545372	1543.46112792	1
11	[0, 2, 1]	3378.30482450	0.0000000	38.4753327	-12.8916085	1646.54479823	-1
12	[1, 1, 2]	3407.65235274	33.5687710	15.6700397	16.9834007	1660.84842756	1
13	[1, 0, -1]	3415.87571821	33.5687710	-3.5676267	-22.9186834	1664.85639615	1
14	[1, -1, -1]	3485.58392475	33.5687710	-22.8052931	-7.2033015	1698.83133057	-1
15	[1, -1, 0]	3642.77706155	33.5687710	-22.8052931	11.3358538	1775.44535896	1
16	[1, 1, 0]	3644.36220158	33.5687710	15.6700397	-20.0949100	1776.21793698	1
17	[0, 1, 3]	4026.08320093	0.0000000	19.2376664	39.9020841	1962.26412242	1
18	[0, 2, 3]	4237.58553061	0.0000000	38.4753327	24.1867022	2065.34779274	-1
19	[1, 0, 2]	4531.91565817	33.5687710	-3.5676267	32.6987826	2208.80072717	-1
20	[1, -1, -2]	4738.76761649	33.5687710	-22.8052931	-25.7424569	2309.61786289	1
21	[1, 2, 2]	4815.50390443	33.5687710	34.9077060	1.2680188	2347.01819895	-1
22	[0, 2, 0]	5064.22971425	0.0000000	38.4753327	-31.4307638	2468.24414202	1
23	[1, -1, 1]	5210.34702687	33.5687710	-22.8052931	29.8750091	2539.45994803	-1
24	[1, 1, 3]	5404.86267112	33.5687710	15.6700397	35.5225560	2634.26451388	-1
25	[1, 2, 1]	5424.22706659	33.5687710	34.9077060	-17.2711366	2643.70248538	1
26	[1, 2, 3]	5617.15757081	33.5687710	34.9077060	19.8071741	2737.73447321	1
27	[1, 0, -2]	5864.61606195	33.5687710	-3.5676267	-41.4578388	2858.34274053	-1
28	[1, 1, -1]	5878.28236879	33.5687710	15.6700397	-38.6340653	2865.00353273	-1
29	[1, -2, -1]	6087.40698842	33.5687710	-42.0429594	8.5120804	2966.92833601	1
30	[1, -2, -2]	6145.03402815	33.5687710	-42.0429594	-10.0270750	2995.01505625	-1
31	[0, 0, 3]	6346.69572838	0.0000000	0.0000000	55.6174660	3093.30252312	-1
32	[0, 1, -2]	6477.92438963	0.0000000	19.2376664	-52.7936926	3157.26178103	-1
33	[0, 2, 4]	6782.79112646	0.0000000	38.4753327	42.7258575	3305.85013103	1
34	[0, 3, 3]	6981.20271741	0.0000000	57.7129991	8.4713203	3402.55353406	1
35	[0, 3, 2]	7041.93060209	0.0000000	57.7129991	-10.0678350	3432.15156852	-1
36	[1, -1, -3]	7402.32813675	33.5687710	-22.8052931	-44.2816122	3607.80495589	-1
37	[1, -2, 0]	7440.15677722	33.5687710	-42.0429594	27.0512357	3626.24217646	-1
38	[1, 2, 0]	7443.32705728	33.5687710	34.9077060	-35.8102919	3627.78733251	-1
39	[1, -2, -3]	7613.03789641	33.5687710	-42.0429594	-28.5662303	3710.50233719	1
40	[1, 0, 3]	7724.68262855	33.5687710	-3.5676267	51.2379379	3764.91662557	1
41	[0, 1, 4]	7766.84544878	0.0000000	19.2376664	58.4412394	3785.46627278	-1
42	[1, 2, 4]	7829.18806571	33.5687710	34.9077060	38.3463294	3815.85130816	-1
43	[0, 2, -1]	8160.53143252	0.0000000	38.4753327	-49.9699191	3977.34404650	-1
44	[1, -1, 2]	8188.29382072	33.5687710	-22.8052931	48.4141645	3990.87509780	1

rank	n	q	k_x	k_y	k_z	kappa_1^2	chi
45	[0, 3, 4]	8330.85166125	0.0000000	57.7129991	27.0104756	4060.35606028	-1
46	[1, 3, 3]	8361.56732762	33.5687710	54.1453724	4.0917922	4075.32650354	-1
47	[0, 3, 1]	8513.03531531	0.0000000	57.7129991	-28.6069904	4149.15016368	1
48	[1, 3, 2]	8755.47031325	33.5687710	54.1453724	-14.4473631	4267.31004135	1
49	[1, 1, 4]	8812.44981802	33.5687710	15.6700397	54.0617113	4295.08116091	1
50	[1, 3, 4]	9378.04117052	33.5687710	54.1453724	22.6309475	4570.74352643	1
51	[2, 0, 0]	9510.04881200	67.1375420	-7.1352534	-8.7590562	4635.08244985	1
52	[1, 1, -2]	9522.57936453	33.5687710	15.6700397	-57.1732207	4641.18968918	1
53	[2, 0, 1]	9548.88702438	67.1375420	-7.1352534	9.7800991	4654.01172352	-1
54	[2, 1, 1]	9620.97294097	67.1375420	12.1024130	-5.9352828	4689.14552498	1
55	[1, 0, -3]	9723.73323421	33.5687710	-3.5676267	-59.9969941	4739.22964560	1
56	[2, 1, 2]	9874.63132987	67.1375420	12.1024130	12.6038726	4812.77554728	-1
57	[1, -3, -2]	10083.41529694	33.5687710	-61.2806258	5.6883069	4914.53432063	1
58	[1, -2, 1]	10203.28339454	33.5687710	-42.0429594	45.5903910	4972.95657760	1
59	[1, -3, -3]	10355.86251320	33.5687710	-61.2806258	-12.8508484	5047.32178950	-1
60	[1, -2, -4]	10491.41859320	33.5687710	-42.0429594	-47.1053856	5113.39017882	-1

Appendix E. Machine-Readable Companion

The companion file `K-Field_recursive_Lucas_Spectrum1785243336.json` contains the complete numerical basis, metrics, 27-node ledger, vertices and edges, recursive closure levels, arbitrary-volume examples, laminar coefficient arrays, 4097-point Fourier samples for $n=3, 9$, and 81 , all 729 residue sectors, the lowest reciprocal modes, energy bands for $m=1$ through 100 , recursive energy points, and cycle-space data.

The JSON schema identifier is `K-Field_recursive_Lucas_spectrum.v1`. All integers are emitted exactly. Floating-point values are emitted as IEEE-754 decimal representations derived from the authoritative basis and the equations stated in this report.

References and Source Record

Source	Use in this report
User-supplied central Lucas Supergroup result and exhaustive aligned volume-filling law	Authoritative recursive census, arbitrary-volume formulas, and exact second-order closure statement
K-Field_Lucas_Supergroup_Analysis_1785208014.pdf	Historical archival format, exact Supergroup census, laminar and reciprocal context
K-Field_Lucas_Supergroup_Complete_Nuclear_Closure_Domain_1785235808.pdf	Historical document design, PI and proprietary classification format, Cell3 hierarchy context
K-Field_current_summary_1784566563.pdf	Canonical reconstructed Cell3 lengths, angles, vectors, face areas, and volume
K_Field_canonical_lexicon_v1.json	Terminology and registered primitive metric constants

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